

10 September 2026

Interview with Industry Partner Mr Harry Lim

Speaker 1: Can you please state your name?

Speaker 2: Harry Lim. Harry.

Speaker 1: What is your involvement with the project on a more macro level?

Speaker 2: Okay. This project is part of a collaborative effort between Air Force, NUS, and my company, IOI Inspire. This project, hopefully, can help to boost the relationship between Air Force and us. So over here, you can take IOI Inspire as the catalyst.

Speaker 1: Understand.

Speaker 2: We understand that many users are actually hesitating to work with IHL—higher...

Speaker 1: Institutes of Higher Learning.

Speaker 2: Institutes of Higher Learning. Reason because IHL typically start with a lot of R&D, very low TRL. Similarly for Air Force. But the good news of why this particular Air Force team wants to work with us is because they are the innovation team of Air Force.

Speaker 1: Okay. So, what else can you share with me about what is the current method the Air Force uses for FOD clearing?

Speaker 2: It's manual. They involve two staff, one of them driving a vehicle. It's actually a pickup truck. And behind the pickup truck, there's actually a space where they can put Speakers and things like this. So one person drives; the vehicle commander will look out for FOD. When they detect FOD, they will stop the vehicle, and then the vehicle commander will step out of the car to pick up the FOD.

That's how they inspect. And there's no fixed route around the runway. It's pretty random; they go left, right, center.

Speaker 1: So there's no clear route that they follow.

Speaker 2: Yeah, there's no fixed route.

Speaker 1: These two staff you mentioned, are they NSF or... any idea?

Speaker 2: I can't recall. NSF, NSF. Yeah. They do inspection, I think, twice or three times a day.

Speaker 3: Three times a day.

Speaker 2: Three times a day.

Speaker 1: Okay.

Speaker 2: You see, that's another value-add if ours is autonomous.

Speaker 1: I see. Three times a day. Okay. At what times do they occur?

Speaker 2: Early morning... I recall...

Speaker 1: Fixed hours, every few hours?

Speaker 2: I think 7:00 AM, is it? Or 6:00 AM.

Speaker 3: 7:00 AM, 2:00 PM, and after 6:00 PM.

Speaker 4: Did I hear they inspect the runway only three times a day and not after every flight?

Speaker 2: No, no, no.

Speaker 1: Is there a reason for these timings? Why only 7:00 AM? Is it like a strong reason why at 7:00 AM that you must check, or is it...

Speaker 2: Okay, they highlighted that they have a team, a crew—a team of crews. So this crew need to be scheduled to go down to the runway to perform the inspection. And then, why three times a day or two times a day, I think it's after some trial and error, and that's the SOP. Exactly why, no idea.

Talking about scheduling, that is another research topic that they came to ask us also. Just to sidetrack, we were talking about scheduling of airfield controllers. Those guys in the control tower. The scheduling of the controllers is also another challenge. Anyway, that's a separate topic.

Speaker 1: Okay, understand. How has FOD affected some of the daily operations for the RSAF? Any idea?

Speaker 2: We asked them this; they didn't give us a direct answer. Because they cannot say... if they ask, like, "Oh, the plane will crash." So they didn't give a direct answer. But from our know-how, no news so far about...

Speaker 1: No news is good news.

Speaker 2: Yeah, no news is good news.

Speaker 1: What are common FODs that are spotted? What did you see on that day, perhaps?

Speaker 2: On that day, it's a small metal piece, about 2 cm.

Speaker 1: Any idea where these metal pieces come from, or is it just no investigation was done as to why there are random metal pieces?

Speaker 2: I believe behind the scenes they will lock down. Okay, but the pieces, they informed us, if I remember, something about nuts and bolts. These are the usual pieces that dropped out, or some scrap metal.

Speaker 1: I'm curious, because aviation uses rivets, right? Why would there be nuts and bolts?

Speaker 2: Rivets... To me, I'm not an engine guy, so to me it's the same! They used the word rivets. But these are some of the things that they mentioned. Again, they are not very specific.

Speaker 1: Scrap metal. Okay. It will be ambiguous on what shape they will be, but we just know it's metal. Small-piece metal, okay. How frequent does FOD become a problem?

Speaker 2: Oh, first, did they mention plastic? Plastic, also.

Speaker 1: Plastic also? But is it also the similar type, like random plastic pieces? Small piece, palm-sized?

Speaker 2: Yeah.

Speaker 1: How frequent does the FOD become a problem? No clue?

Speaker 2: No clue. They refuse to answer that.

Speaker 1: Maybe also ops-sec. What are some events where FOD has become a problem in our history? Any idea? No one?

Speaker 2: No, right? But FOD, as I mentioned for SIA, happened at Taoyuan Airport in Taiwan. But that FOD is not FOD; that's actually a service vehicle that happened to be parked in the wrong lane, and then the plane just crashed into it. Just in case you're not aware, that was the happening. So if we can detect foreign objects—it could be a car, too. And that type of incident is even more critical because it's human error, and human error definitely will happen.

Speaker 1: Yeah. In what ways do you think an autonomous solution would be helpful in your current operations? You mentioned about the two staff driving, then must look out for the vehicle, so there's some accuracy issues, timing, human fatigue. In what ways would an autonomous solution be helpful to the current operation?

Speaker 2: Autonomous vehicle can inspect the runway more frequently. Autonomous vehicle, if it has an optimum path, pre-designed path, then the inspection can be faster, meaning the frequency of inspection also can increase. Autonomous vehicle means servicemen can be released for other critical roles, given that the armed forces is shrinking due to lower birth rate.

Speaker 1: Do you know how much damage FOD costs the Air Force? X amount of dollars?

Speaker 2: This is actually very important, because no matter what we propose, we want the ROI, right?

Speaker 1: Correct. Need to have some figures to give them a figure.

Speaker 2: But unfortunately, they cannot tell you numbers. And then the servicemen that we talked to are not of that level, either.

Speaker 1: Yeah, they are just following instructions. NSF.

Speaker 2: Yeah, they just follow instructions, NSF. Correct.

Speaker 1: So I'm assuming manpower costs would just be your NSF price, NSman price.

Speaker 2: Okay, that's the interestingness of Air Force. Actually, budget is never the issue. Because NSF basically work almost for free. The market rate is so much lower compared to a commercial rate. It's the technology itself. Whatever technology they are going to deploy, there's a certain fixture cost and recurring maintenance costs.

Speaker 1: Did the NSFs share with you what are some of the issues and pains that they feel as they are doing their routine inspection? Did you talk to them?

Speaker 2: They are actually quite okay. Just sit in the vehicle, enjoying chasing the birds!

Speaker 1: We got some from other people. Is there any target removal rate or success rate? Or is it 100%? Like, aim to be 100% removal? Let's say off 10, maybe you can pick up nine.

Speaker 2: This target, I think it's more between us and competitors. Because what we need to present to Air Force is the percentage accuracy and thoroughness. So in our piloting, we will put maybe 10 objects or 100 objects, and then we see whether our

technology can comb and then identify all the 100, for example. And then that will give us a number.

And then Air Force will take that, and when we are ready for commercialization, then they will raise a tender. So Air Force itself, like many of the end users, they don't have real, clear visibility on what are the thresholds or targets; they don't have. It's always a call for proposal, and then based on that, average out, and then issue a tender that's reasonable.

Speaker 1: I have also another question. You mentioned they use pickup truck, right? So pickup truck got quite a big volume, like the boot. So what is the expected volume that they are supposed to pick up in one run?

Speaker 2: Very, very minimum. Very minimum.

Speaker 1: Would you say it's less than one handful?

Speaker 2: Yeah, it's like one palm. The day when we were there, they picked up something that is even smaller than 2 cm. So very, very minimum. But they mentioned that they do pick up FOD. So I suspect if for X days they don't bother to pick up, right, that can be a disaster already.

Oh yeah, you need to include bird shit, also.

Speaker 1: Bird shit or something?

Speaker 2: Must filter that out. Must know there's bird shit so that it doesn't go in a false positive.

Speaker 1: Anyone else has any questions?

Speaker 5: I'm good.

Speaker 1: All straight, all straight.

Speaker 2: Any question that you want me to present to Air Force?

Speaker 1: I think cost is the key thing. Because if they want us to kind of give them a good ROI number on the interim...

Speaker 2: It's too early. They know, they know.

Speaker 1: They know that, okay.

Speaker 2: I think it's more of what are we already contributing. And then they don't cause you a pai seh, right?

Speaker 1: So what I think we can do is that we can kind of estimate how much it costs to run our solution. Then, I don't know, in relation to how much they spend, like what that figure looks like.

Speaker 2: If you can, budget always good to secure more.

Speaker 4: It might not be a purely cost thing, too. I mean, I think cost is important, obviously, but it might be, you know, if they could do it autonomously as opposed to having to use manpower, that might be a selling factor as well. Even if it costs slightly more, I would say. But yeah, if you can say it's cheaper, obviously the goal is to be cheaper, for sure.

Speaker 1: Yeah, for sure.

Speaker 2: We have the advantage of a local. Just that once we really commercialize, right, this localness doesn't matter anymore.

Speaker 1: There's another thing that I'm really interested to know: what other solutions have they ever tried before? Are we really the first solution that they are trying, the very first?

Speaker 2: They look up to Changi, Changi Airport.

Speaker 1: So their practices, what they are currently doing...

Speaker 2: So currently they are sharing the same runway at Changi. But in one to two years' time, there will be a new runway which Changi will use, or Changi will take over the current runway, then the Air Force will use the new runway. I think that's going to happen in two years' time.

Speaker 1: So currently they have not tried other solutions for FOD?

Speaker 2: I believe they should be already calling for latest state-of-the-art. Just that they want to know whether we can do something or not. But realistically, I look at it, it's going to be a challenge for us. Because I know innovation is not something that can happen by tomorrow; it will take years, especially safety is concerned.

For us, we can look at this and then create a pipeline so that maybe after two, three years, or five years of iteration with different teams, finally NUS has another technology that's ready. Some of the projects that Prof and myself have worked on, we spent like eight years.

Speaker 1: 2018.

Speaker 2: Yeah, you see so much white hair already! From like you, all the way to white hair. But not to be disheartened nor disappointed, because this innovation is something

that you need to experience yourself. No one can teach you; you need to experience. We can only create the framework for you to work in. Then once you work in, you understand what to balance, what to prioritize, what to juggle. The next round when you are out in the market working, you know what to do and what to expect. Or if you want to create your own company, your own startup, this experience will help.

Speaker 1: What do you think also is how fast do they drive, like how fast do they complete one round of clearing?

Speaker 2: Okay, we asked them a few times about this. They mentioned 40 km/h.

Speaker 1: 40 km/h?

Speaker 2: Over the whole airstrip of six clicks. So looking at the randomness, the way they drive, they just want to get the job done fast.

Speaker 1: Because I would think, if I was NSF, why would I only drive at 40 km/h? It's very slow. I would go faster.

Speaker 2: Because they need to eyeball. So I asked them, "You cannot drive too fast, also, right? Then you will miss all the FOD." So they say, "Yeah, yeah, 40 km/h."

40 km/h to me is a bit fast already. We were quite fast, you know. Then I was surprised that they finally stopped and then pick up something.

Speaker 4: How they do it now, just manual?

Speaker 2: Now manual eyeballing.

Speaker 4: There's got to be some, just for accuracy.

Speaker 1: Definitely for accuracy.

Speaker 2: 40 clicks is actually very fast already.

Speaker 3: We don't even know how accurately they... how thorough they are.

Speaker 2: They couldn't answer us. They said, "Hey, see, we pick up something!"

Speaker 1: I'm guessing they didn't have any way to verify. They don't know what is the total number also, so they also won't know how much they can pick up is the total number.

Speaker 2: See, that's why even if we go down and meet them, right, I tell you they can't tell you anything. Prof and me have gone down on different projects, and this is what we observed. Most of the end users are truly end users; they just follow SOP. It's a 9-to-5 job for them; don't come and disturb them after that.

We were talking about checking of surgical tools. They just eyeball, move, eyeball, move. They make mistakes. We have gone down to a hospital and pointed out the mistakes that they make.

Speaker 4: I wonder if there would be any data, any assumptions that could be made about how accurate a human...

Speaker 2: At 40 km/h?

Speaker 4: Yeah, if a human eye can see as good as a drone.

Speaker 1: Probably if reflective thing, you can see very fast because it's very obvious, like, something blinking, okay, you can identify. Anything that doesn't have that kind of property...

Speaker 4: They probably never measured it, I would imagine, right? I don't think they know.

Speaker 2: Then I also asked more questions. I asked, "What is the minimum size that you're looking for?" Then he picked... he said, "Something like this."

So for them, of course, they want something that's super, super accurate at the cheapest cost. Chinese say, *you pian, you cheng, you hou*. Meaning must be very good, must be very cheap. Something like this, Hokkien saying. So all users are all like this.

Speaker 4: That's another value-add, probably, right? If they have a minimum size that they actually look for, and you can get it even smaller than that with the same amount of... you can argue that as an **improvement of the process as well.**

Speaker 1: Yes.

Speaker 2: But I think for us, what we can do is literature survey: what have commercial companies committed commercially? Because commercial commitment is something they have to abide to. And if any accident happened, they will get it. This particular small litter, 0.05 cm rivet, blow out a plane upon landing or blow out a wheel, then that company will be at stake for suing. So this is something that we need to look at. We can find what commercial companies are committing, then those will be the benchmarks that we measure against, not what Air Force is saying. I don't trust this team.

Speaker 1: Any other questions? What about drones? Anyone ask about drones?

Speaker 3: Drones, other than the height limit, would there be any...

Speaker 2: We will definitely need clearance. Need to check the CAAS.

Speaker 3: And also what are the restrictions, because we are planning to put spotlight under the drone.

Speaker 2: That's okay. It's more of the drone falling down. That's why CAAS dictate that the drone must have a tether at certain zones of Singapore, especially the zones around the airports, critical areas. So I think, if I'm not mistaken, only maybe one-third of Singapore or lesser can have drones flying beyond 50 meters or 200 meters. Very limited. So Singapore is actually a very difficult place for drones.

When I was running the Smart City Lab for CapitaLand, my understanding is that a lot of drone application is actually from the coast to the sea, to the ships, and so on. Shipment of medical supply or something like this. So drone within Singapore, very little use cases. Singapore has a few drone startups, and they have not seen success.

Speaker 4: We've had drone projects here, and even the projects have had to deal with that issue of how high they can fly and stuff. There was a guy named Brian that used to work here; I think he was the drone guy.

Speaker 2: When I was running an incubator for PUB, drones are used for inspection of tunnels. But that's okay. Then the challenge is the tunnels are of varying diameter. Then PUB spec that the diameter must be very, very small, something like this. I can't remember, maybe it's like a 20 cm or lesser. So the drone must be very, very small, flying inside the tunnel. That's also another challenge.

Speaker 1: Battery life.

Speaker 2: And the drone inspecting in the tunnel definitely tethered. Definitely tethered.

Speaker 1: That's all for my questions. Anyone else?

Speaker 2: No worries. Just keep...

16 September 2026

Interview with 3SG(NS) Prady, Paya Lebar Air Base

Interviewer: Just for the record, can you tell me what's your name, and like, your rank, and what's your role in the whole process?

Prady: So, my name's Prady. You can call me Prady, because I don't like using my full name. I was a **third sergeant, air force technician**, and I was in the fighter series squadron. So that's ground systems, like arrestor wire and stuff. **And then I had to clean FOD also.**

Interviewer: So, was like, wasn't it like a part of your job, or like is it the only job that you do?

Prady: **It's a part of my job.** Because after we work, we have to clean, and, um, at the times that we work, the, the people who drive the FOD vehicle are not like in base. So we have to take a broom and...

Interviewer: Oh my god! So you guys take a broom and you just walk the, the...

Prady: I can't, I can't specify how long, it's ballpark of 4 kilometers, **but it's around there.** **Walk and broom. Walk and broom. And the broom is like that [expletive] broom.**

Interviewer: Like a normal broom?

Prady: Jus a normal broom. In fact, it makes more FOD than it cleans up.

Interviewer: Oh dear, okay, okay. Okay. So, in your experience, like how, how often do you like deal with FOD, like let's say in a week?

Prady: **Oh, every day. Every day you need to do.**

Interviewer: Okay, wait, maybe take me through like, what do you do on a daily operational-like basis? And then like, what, how does FOD like fit into that picture?

Prady: Okay. So, uh, for our ops, it's generally like, uh, so we got the daily checks on all our systems. So, we'll go like, lights, lights are one of our systems. Then if, if there was a storm, then sometimes there's lightning strike, right? **Then, uh, the runway will crack, and then there's like rocks that we need to pick up. And sometimes the lights will also crack, so the glass also need to clean up.**

Interviewer: Mhm.

Prady: Then there's, uh, there's like monthly, four-monthly, and yearly servicing.

Interviewer: Okay.

Prady: Those are bigger. So, so, uh, so wait, let me show you a photo of the arrestor gear. Uh... It's like a very massive machine, right? Then we need to open the whole thing on the runway. Is this thing? We need to open the whole machine on the runway and, like, take the gears and all the parts out. Let's put this way. Uh, take all the parts out, and then, and then fix inside, and then put everything back together. Right? After you've put it back together, right, sometimes you'll find like a screw here, a screw there, right? Then you have to clean that up. You have to do a walk around the area to make sure that nothing dropped. Because if it goes into the intake, then you're [expletive].

Interviewer: Something that I'm, I'm quite curious about is that in aviation you guys use like a lot of rivets traditionally. So, why would there be like screws? Or is it different from...

Prady: Oh, uh, so for these ones, right, cause vibration's not as big a factor. Mhm. There's a lot of screws. We just, we just locktight and screw it in.

Interviewer: I see, okay. Got it, got it.

Interviewer: So is it, is it every day annoyance type?

Prady: Yes, yes.

Interviewer: Okay. So, like, what are the current methods that you guys adopt, other than, I mean what you just shared earlier, like using the broom, walking the air strip, or...

Prady: Okay, so officially the method is, uh, there are these, I think four or five cleaning trucks, right? They will go and sweep the entire... It's like those road-sweeping trucks. Mhm.

Interviewer: So, is it, it's like a specialized vehicle?

Prady: Yeah, it's a specialized vehicle. They will, they will do for two hours every morning and two hours every night before, before and after flying.

Interviewer: Mhm.

Prady: Yeah. And like, they are like pickup trucks with like modified bags or something?

Prady: Oh, no, it's, it's, it's just a, it's a, wait, let me see if I can find a picture for you. It's a specialized truck that is only for FOD.

Interviewer: Mhm.

Prady: Okay, this is not it. Uh, similar to a, it sweeps the road, right? Similar, similar like that. It's, it's just a lot bigger.

Interviewer: Got it. Okay. So, can you share me like what's the like, logistics or manpower needed in that process?

Prady: Uh, yeah. So, it's, those, those are all contracted out from, I think from Changi Airport or somewhere. They, they got one contractor who comes in to do it.

Interviewer: Okay.

Prady: So, it's not really on, not really our side.

Prady: But, but again, once they are off duty, then anything happens, the [expletive] is on us. And we don't have... So what ends up happening, right, is when they are off duty, uh, we will take a, we'll either take a tractor, uh, which is like the, the towing tractor for the aircraft, right? Or we'll take our normal, uh, D-Max car, and turn the lights all the way to the max, and then just, walk, walk beside the car and watch, watch and see if we can see anything on the floor, then we'll broom it up.

Interviewer: So, it's like walking speed?

Prady: Yeah, pretty much. But, do you have to do it for the entire strip?

Prady: Yes. I, I could, I could imagine that, that takes quite long, like upwards of, you know, an hour plus.

Prady: Horrible. Horrible, horrible. Very manual process, right?

Interviewer: Okay. So, to your knowledge, right, is there any other method that, like, you guys have, air force has tried to, to innovate in this area?

Prady: Uh, I think, I think we tried to use a vacuum cleaner once, but it didn't work.

Interviewer: Vacuum cleaner?

Prady: Yeah.

Interviewer: Like, a manual vacuum cleaner?

Prady: The hand-held, hand-held ones.

Interviewer: Hand-held ones, okay. Why, why did it not work? Was it like...

Prady: It's just, again, like if you're walking and brooming, it's the same as vacuuming and brooming. You're still walking, right?

Interviewer: Correct, speed is the, yeah.

Prady: Yeah. It's, it's still slow and you still have to do manual work. Mhm. Uh, I mean, someone tried to tie a vacuum cleaner to the car, but that also doesn't work, right? It's, it's kind of...

Interviewer: Jenks, jenks, stinks.

Prady: Okay, what, what are some common like FODs that you guys like come across every day? What do you encounter the most?

Prady: So, uh, screws, rivets. Screws, rivets are quite common. Uh, rocks from the... Rocks, screws, rivets. Uh, actually, bones are quite common.

Interviewer: Bones?

Prady: Yeah. So, like rats, sometimes, right, there are a lot of rats on the runway, then, then it will die on the, die on the thing, then it will get roasted by the engine or what, right? Then after that, you just see the bones of the rat over there.

Interviewer: Oh, like literally like evaporate the...

Prady: Yeah. Okay.

Prady: Uh, then, yeah, then glass, glass is quite common. Glass? Shattered glass.

Interviewer: Ah, okay, interesting.

Prady: Oh my god.

Interviewer: Okay, okay, okay. What, what, what, what we say is like the volume that you pick up in like one, two-hour session?

Prady: I mean, I mean, um, maybe a handful.

Interviewer: A handful?

Prady: Yeah. It's, it's not too much, because the sweepers take care of most of it. Mhm. But like the manual one is probably like just a handful.

Interviewer: Handful. Okay. There are, okay, wait, which air base are you all working with? Do you know?

Prady: Probably Changi Airport.

Interviewer: Changi Airport, yeah. Okay, then that's, then it'll probably not be as bad, because Paya Lebar Air Base, right, is really [expletive] old, so the pavement is cracked.

Interviewer: I see.

Prady: And then the pavement keeps coming up out, so we need to carry, like, I've carried like a piece of pavement this big, out from the... The entire, the whole pavement, the whole pavement cracked, and then, uh, there was a C-17 landing that day, it landed, and because of the vibration, the whole pavement just came out, and then you have to carry it out.

Interviewer: Oh dear.

Prady: Yeah, that is, that's special.

Interviewer: SAF, right?

Prady: Okay.

Interviewer: SAF, SAF things. Yeah.

Interviewer: Okay. Uh, do you think there'll be any friction for the air force to adopt a new process in order to clear FOD? Like, example for what you guys are doing now is quite like, like, mandated. Like, you guys have learned and tried it a lot of times. So, if you introduce a new system, do you think there'll be friction in learning the new...

Prady: Uh, no, I don't think so. Okay, so in my experience, right, when it comes to new technologies, the RSAF adopt very, very fast. Mhm. Some of the regulars are a bit, um, you know, traditional.

Interviewer: Yeah.

Prady: And it takes them, like, they, they need to see that it works better than their method, otherwise they will not adopt.

Interviewer: Understand. Okay. Do you know how much FOD costs the air force? Like in damages. Have you ever seen a FOD causing damage to some, some, some of value?

Prady: Okay, I, I cannot, obviously I cannot share exacts, but I have like experiences. When, when we, when we do our training, they show us like videos of where a rivet has gone inside a, inside the air intake, and then they have to clean the entire, fix the entire... So, I think it was for, if I'm not wrong, for the C-130, it went into the prop, and then, and then they have to replace the entire thing.

Interviewer: Wow.

Prady: Yeah.

Interviewer: And there was one rivet that caused that damage.

Prady: Yes.

Interviewer: Okay, got it, got it. Okay. Uh, okay. So, can you share me like you are just one person. So, I guess as a sergeant you have a few men that with you, right?

Prady: Yeah.

Interviewer: So, do you think it's a manpower issue in terms of like collecting FOD? Let's say if you were to put like maybe like two-section, three-section of people to do it, will it be faster in that way?

Prady: Uh, I mean, how many people can, can really, like, help? Because it's, we've got a, about four kilometers of runway, right? Okay, so, so there's this thing called RSAF safety day, right? Every Thursday, every first Thursday of the month, uh, they talk about safety and all. Mhm. Then on that day, the entire air base goes down to do FOD walk together.

Interviewer: Mhm.

Prady: And even then it takes us one hour to clean. So, I mean, I mean, no need to say already, there's a lot of [expletive] people on the runway.

Interviewer: Yeah. That's the entire, the entire RSAF walking on the... And it still takes one hour to clear.

Interviewer: Okay.

Interviewer: Okay, anything? Because this large operation is, what do you think about some of the existing solutions that are already out there in other countries, such as friction drag mats, friction drag mats and specialized vehicles? I mean, you already mentioned that you're using. Yeah. Do you think they boost productivity in general?

Prady: I think, I think the, the specialized vehicles help us a lot. Cause, uh, I mean, we, okay. Can this be off the record?

Interviewer: Off, off as well.

Prady: Okay, so, uh, this one, okay, so we, obviously we do our cleaning up, we do due diligence, but we are not as scared that we leave something behind, because we know in the morning the specialized vehicle got to come and sweep again. So, okay.

Interviewer: Okay, yeah, get you.

Prady: So, I guess they are the first tier to clear, and then you guys are second tier.

Prady: Yeah, second tier.

Interviewer: So, is it because, like, first and foremost, we may not know how many FODs there are on the entire runway?

Prady: Yeah.

Interviewer: Then they clear most of it.

Prady: They clear most of it, but then the thing is they are inside the vehicle, right? They cannot see what's on the ground.

Interviewer: Yes.

Prady: The people on the ground are the only ones who can verify if it's on the ground or not. If it's a threat or anything.

Interviewer: Okay, understood.

Interviewer: Okay, very cool. Yeah, so we are, we're working on a research study/stress-like product concept. It's our FYP la. It's our 4301, is to have a drone and a UGV work together, to have it like scan and see such a small piece, and then like give instructions to the UGV to then collect such a FOD. But, will, are we, are we guessing that like, um, for bigger pieces, of course cannot la.

Prady: Cannot la.

Interviewer: Yeah, but definitely for smaller pieces it will be very, very helpful.

Prady: No, it's the smaller pieces which we need help with, because the big, the big sweeper does not pick up the small pieces. It will always miss the small pieces.

Interviewer: Right.

Prady: The big pieces, I think they, like the bones and all, right, it settles on its own.

Interviewer: Mhm.

Prady: But I think the smaller stuff, right, like rivets, the ones that can actually cause damage, sometimes the sweeper will not pick it up.

Interviewer: I see. I'm wondering why it cannot pick it up, because I'm guessing they were designed for it, but unable to address it.

Prady: I also think that, off the record la, the contractors don't really do their job properly.

Interviewer: I see. Yeah, I mean, human error, whatever error can happen, yeah. Got it, got it.

Interviewer: Okay. Oh, we already use drones for some ops, like, for example, right, I was telling you, uh, the pavement, so like, uh, potholes and stuff, right, we use drones to identify size and, uh, like urgency to clean up or what.

Interviewer: Okay, sick. Sick, sick, sick.

Prady: But it's the one massive drone, it's like the really big one.

Interviewer: Ah.

Prady: Then it goes up, then it comes back down. The problem with drones, right, especially in air base, is you need to take special permission every time you fly it.

Interviewer: Correct, correct, yeah.

Prady: Cause you need to turn off the air space while, while you're flying it.

Interviewer: So, right, I understood, I understood it.

Interviewer: So, how, how often does that drone process take place in your...

Prady: Bro, we, it got to a point that the paperwork to get it on takes so long that we just, once, once every couple months just to show for, for publicity, we'll fly it.

Interviewer: [Laughs]

Prady: And then other than that we'll just manually do it ourselves.

Interviewer: I see. Okay, okay. Yeah. I, do you know Changi Airport has a thing called iFerret? Have you heard of it?

Prady: I don't know about it.

Interviewer: Basically, it's like a, it's like a super camera that can, like, spot exactly where on the runway there's a FOD. And then it tells, like, you know, it tells, um, somebody tells somebody to, can you go and clean it up at this place?

Prady: Oh, that's pretty sick.

Interviewer: Yeah, but there's, there's only at Changi Airport la. And it's only at one runway at Changi Airport.

Prady: I wish we had that.

Interviewer: Yeah, pretty crazy, yeah.

Prady: I have an idea for you la.

Interviewer: Yeah, sure, sure, come.

Prady: You know those, um, 3D scanners? Like the very high precision...

Interviewer: We tried to use those also.

Prady: 3D scanners. So, it's a very expensive lidar. You pair with like computer vision and like cameras, right? So you, you will have like, like along the runway you have markers to like mark the, the, like a single piece of like maybe a section of the runway. Then you create a 3D scan of the runway, uh, without any FOD. So, like, it can map out, like, the contours, no, the contours of like the small rocks and stuff. So you can, like, then it gives you the additional camera, anything additional. But you need a very high precision.

Interviewer: Correct, correct.

Prady: But we tried to do that, you know?

Interviewer: Really?

Prady: But not for FOD. We tried to do it for pothole.

Interviewer: Uh-huh.

Prady: Then we looked at the quotation for the 3D scanner, right? \$50,000. Then I was like, "Okay, you know what?"

Interviewer: That's pretty crazy.

Prady: But then, but then RSAF, right, it's not like NUS that you have a limit. They go and buy already.

Interviewer: Oh, serious?

Prady: Yeah. Now, but then, by the time they buy, I don't know, I don't know what they use it for now. But they have it la.

Interviewer: Okay.

Interviewer: Cool, cool, cool, cool.

Interviewer: Thanks. Okay, is, is this been very, very insightful, very, very, very.

Prady: Is this all record now?

Interviewer: Yeah, that's all.

Prady: Okay. One of the other things that we do for FOD, right, so runways are around grass patch, right? We will take the leaf blower and just blow it into the grass and just [expletive] get out of there.

Interviewer: Ah.

Prady: Wait, technically it's okay, because not on the runway, right?

Prady: Once it's off the runway, don't care.

Interviewer: But if it's still close to the runway, it can still be sucked in, so...

Interviewer: I see. Okay, okay. Well, that kind of makes sense though, I guess.

Interviewer: Yeah.

Interviewer: Cool, cool.

Interviewer: Cool, cool.

Interviewer: Okay. Thanks, thanks Prady for your input.

Prady: No worries.

Interviewer: Yeah.

Prady: I'm, I'm, I'm all supportive of this [expletive]. Like, I know even in reserves I probably won't have to do this shit, but...

Interview with LTA(NS) Elnathan, Tengah Airbase

16 September 2026

Speaker 1: Yeah. Maybe I introduce myself first, before I can introduce yourself. So, my name is Asuka. I'm Josh's friend. I met him on exchange, actually. Also realized that he's an Air Force officer in the same batch that I was in OCS. So, I guess it must be the same batch at SIM, probably.

Speaker 2: Yes, yes, you're right.

Speaker 1: Yeah.

Speaker 2: Yeah.

Speaker 1: So, basically right now I'm in my final year in NUS, and I'm doing my final year project. And what we are doing is that we are trying to kind of apply drone to bot communication technology in some sectors that we are interested in. So, one of them is actually FOD clearing on runways, because we feel that we can take advantage of having an aerial view using the drone and communicating to the bot to speed up the process. I understand from Josh that you're involved in the process. To what extent, I'm not really sure. So, maybe you can introduce yourself and also what is your background in it.

Speaker 2: Yes. So, I was the air traffic controller in Tengah Air Base. So, yes, FOD, that's what we do.

Speaker 1: Haha.

Speaker 2: So, I used to inspect the runways to check for it, and yes, we also have drone operations to specifically check for stuff on the runway, like visually check. It's quite crude, but they just want to say on paper that you can do it.

Speaker 1: Mhm.

Speaker 2: For the drone operations, since it's more relevant to you, they have I forget, some US brand of drone. They don't want to use DJI because it's Chinese.

Speaker 1: Yeah, makes sense. Understand. By the way, I just want to thank you for taking the time to talk to me today. I think it only takes about 15 to 20 minutes maximum. Don't worry about it.

Speaker 2: Don't worry.

Speaker 1: Don't worry.

Speaker 2: Yeah.

Speaker 1: Okay. So, you're saying like they already use drones to scan the runway for?

Speaker 2: Yeah, so the drone has a set course plotted already. So, you just take off from the tower, then it goes up and down the runway to get everything. You have to see visually. There's no indicators that pop up. So, you have to actually see if there's anything on the floor, and the drone comes back. So, that's pretty much the extent of how we use drones for runway inspections for FODs, or any cracks or whatever. It's not used that often. It's probably used like once or twice a month. So, mostly it's just the car, driver, a car and you, the officer are actually going up and down looking at the roads. Yes, so, yes, that's how we do FOD pretty much.

Speaker 1: Cool. Can you share with me, like, some of the

Speaker 2: This was 4 years ago?

Speaker 1: Two years ago.

Speaker 2: Two years ago.

Speaker 1: Okay, maybe times have changed, I don't know. But okay, can you share with me some of the process, the time that it takes, the logistics and manpower that you need in this process of clearing FOD, with what you're familiar with, at least?

Speaker 2: Like, in general when you find one or using the drone, or I think more of just what you currently do. Like you mentioned the pickup truck driving around, so how did that work?

Speaker 1: Okay.

Speaker 2: Okay, so we have like first inspection, morning inspection, the first light of day. Usually start around 7. On paper, that's when the sun rises, but not really.

Speaker 1: Mhm.

Speaker 2: So, the driver, the officer, go into the truck, and they'll drive up and Okay, so besides the runway, let me find It's not hidden at all. Tengah Air Base Google Maps. I'll show you the thing.

Speaker 1: Of course. Yeah.

Speaker 2: Okay, so this is the whole runway.

Speaker 1: Is this Tengah?

Speaker 2: Yeah. Looks like it.

Speaker 1: Yeah.

Speaker 2: So, where's the mi- Okay, this is bringing back so many memories. I remember this. Where is the start?

Speaker 1: Mhm.

Speaker 2: Okay, here's the tower. Here's the control tower. Wow, you can see it so damn clearly. Okay, this is the control tower. This is where we'll start off in the morning. Can you see my cursor?

Speaker 1: Yeah, I can.

Speaker 2: Okay. So, we'd start here. That's an inspection route, which I don't remember that well, but we'll go down. Unto this point it still doesn't matter.

Speaker 1: Mhm.

Speaker 2: This is still not within the aircraft maneuvering area, so FOD's not that important here. It's just like a normal road.

Speaker 1: Yeah.

Speaker 2: Okay, we go in somewhere

Speaker 1: The taxiway is it?

Speaker 2: Somewhere around here, I think. Actually, no, this is Aircraft already here. I'll just briefly run through. It's not correct anymore. But okay.

Speaker 1: Mhm.

Speaker 2: Shoot, sometimes kind of big. Okay, we go in around here. Then we What else? We come to this point that's like an FOD checkpoint. You get off, you and the driver go around in the car, you check the tires, check if there's anything loose on the car, check if anything stuck to the tires, nails, whatever. I've found nails on our tires before.

Speaker 1: Nails, like, nails?

Speaker 2: Nails, like, we all nails. Yeah, we take it off, we report, we use the walkie-talkie to tell tower if we found anything. Most of the time there's nothing. Okay, let me just get back in the car, then we'll just start driving. We drive, we drive down. So, anything past an FOD checkpoint, that's where aircraft maneuver around. That's why FOD's important. So,

we go down. This is the taxiway. Aircraft can take off of it, like fighter jets, but they don't do that as often.

Speaker 1: Mhm.

Speaker 2: Go down, we check, go up, the whole runway itself. We'll have to go down, we check the other taxiway.

Speaker 1: Okay.

Speaker 2: Besides the taxiway and runway itself, we also check these loops. One, two, I think it's here. Three,

Speaker 1: Mhm.

Speaker 2: and four. So, anywhere the aircraft might move to?

Speaker 1: Yeah, cuz they don't get parked right out in the open for security reasons. They get parked elsewhere in these sheltered regions.

Speaker 2: Yeah.

Speaker 1: So, if the aircraft goes there, they would

Speaker 2: check it, that part would need to be inspected by the inspection officers.

Speaker 1: Mhm.

Speaker 2: When we do the drone inspections, it's just on taxiway and runways. It's up and down this whole strip. It doesn't go in here. Okay, cuz this spans, I don't remember, quite a few kilometers.

Speaker 1: Mhm.

Speaker 2: Like, the runway itself are like pretty 9,750 feet, I don't remember, something like that.

Speaker 1: Mhm.

Speaker 2: Could I have it on somewhere. But yes. So, the drone inspection is just for this strip. We might extend it a little bit over here. Why? Because there are some lights here, the approach lights that the aircraft sees at night. The drone would see if the lights are all on. Yeah, that is pretty much it for FOD

Speaker 1: Okay, got you, got you.

Speaker 2: in terms of checking. Yeah.

Speaker 1: I see. So, how long does this process take for you every morning? Like, this is a ballpark figure, I guess, like two hours, four hours?

Speaker 2: No, no, no. It's probably like less than an hour.

Speaker 1: Less than an hour? Okay. So, do you think it's because there are many things to check, or it's just because like there's not like this is the process? Why is it less than an hour?

Speaker 2: I guess there is a speed limit,

Speaker 1: Mhm.

Speaker 2: but that being said, if there was no speed limit, can you really tell if stuff is on the floor?

Speaker 1: Yes, get you, get you.

Speaker 2: The thing is there are a few checkpoints we need to get off to check and get back in.

Speaker 1: Mhm.

Speaker 2: The space is kind of large. It's like loops in here there.

Speaker 1: Yeah.

Speaker 2: That's about the reasons why it might be slow per se.

Speaker 1: Right, got you. Okay. So, what would you say is the volume of the things that you collect? Like, let's say on a random day, how often do you pick up stuff?

Speaker 2: Yeah.

Speaker 1: Not often.

Speaker 2: Not often?

Speaker 1: Not often?

Speaker 2: Not often. All right, get you, get you.

Speaker 1: Cuz for anyone working on the runway or whatever, they have extra protocols and processes just to check if they've left anything behind, if they have accounted for all their tools. Specifically because they're working where the aircraft maneuver.

Speaker 2: Yeah.

Speaker 1: All right, get you.

Speaker 2: It's quite rare.

Speaker 1: I see. So, it's a rare thing. So, personally, have you encountered any?

Speaker 2: Yeah, I have. Like, there might be something that flew over, like some plastic, or wood, something that's able to be blown by the wind.

Speaker 1: I see.

Speaker 2: There have been dead birds, I guess.

Speaker 1: Dead birds, like a carcass?

Speaker 2: Yes.

Speaker 1: Oh, damn.

Speaker 2: Okay. That depends. That'll be serious if it was an air strike. It won't be that serious if it's just a dead bird.

Speaker 1: Oh, wait, why would it be serious if it's an air strike?

Speaker 2: Air strike means that the bird either hit the aircraft, that's still okay. But if the bird got sucked into the turbine, that would be an issue because they would need to dismantle the aircraft, check everything in the engine if it's okay.

Speaker 1: Ah, okay. Mhm.

Speaker 2: So, like, if you see a bird with a missing head,

Speaker 1: It's fine.

Speaker 2: it might be an air strike.

Speaker 1: Oh, okay. Okay.

Speaker 2: Okay, that's terrible.

Speaker 1: Okay, that's terrible.

Speaker 2: All right.

Speaker 1: What other things?

Speaker 2: Dead animals show up here and there. In terms of big metal tools or maybe something the size of my fist or a nut this big, we don't really find that.

Speaker 1: Mhm.

Speaker 2: If we do find that it's quite bad, cuz I think each tool has a serial number.

Speaker 1: Yeah.

Speaker 2: And if you find it, some other squadron's going to get in trouble.

Speaker 1: I see. Okay.

Speaker 2: That's I haven't found that before, so yeah.

Speaker 1: All right, get you. Are you aware if there are solutions that other countries have tried, such as specialized vehicles that they use in the US, as well as this thing called a friction drag mat?

Speaker 2: Nah.

Speaker 1: No idea?

Speaker 2: Nope.

Speaker 1: Okay, the next question would be, if, by any chance one day the Air Force adopts them, do you think they kind of boost productivity in that sense?

Speaker 2: For FOD specifically?

Speaker 1: Yeah.

Speaker 2: I feel like FOD in general doesn't take out that much time.

Speaker 1: Mhm.

Speaker 2: But there are times where it could be useful. Sometimes there's a midday check, like in the middle of the day, there's another kind of a full check. That could save time, they could squeeze in more sorties. In the event of a crash, I'm sure you've read about it back in 2020

Speaker 1: Mhm.

Speaker 2: or the F-16 crash. Oh, straight from a movie, I tell you. But okay, if that were to happen, let's say all these were in place, it could play a huge part into making the runway operational again, like, saving time.

Speaker 1: Yeah.

Speaker 2: Yeah.

Speaker 1: Okay, thanks. Thanks for the input. I think I have asked all the questions I need to ask. I just I guess I would just ask for any comments of What do you think about the idea that I've been sharing with you? Yeah, so our idea was basically having the drone and the bot communicate so that they can kind of pick up FOD. And if it's there, then you will go and pick it up. And if it's not, then you don't have to pick it up in that sense. Yeah.

Speaker 2: This was for a final year project?

Speaker 1: Yeah, it's a final year project, yeah.

Speaker 2: Okay, okay. I think any idea can be thrown to the Air Force. They are quite open to getting new ideas.

Speaker 1: Mhm.

Speaker 2: I don't know if they've ever thought about this. They might have, I don't know.

Speaker 1: Yeah.

Speaker 2: They might have, they might not. The idea itself, honestly, anything that saves time and if it's reasonable enough, price-wise and time-wise to implement, they wouldn't say no to. But it would take a while to use it properly, regularly, to replace, in a sense, replace a person driving around.

Speaker 1: Yeah.

Speaker 2: Yeah.

Speaker 1: Okay, get you.

Speaker 2: But like I said, in the event of an emergency or an accident, you can even quote the because it was published as a public thing, the F-16 crash, you can even mention that, oh, if this were to happen, our bots, however, our process would greatly reduce the time it would take for the air base to be operational again, runway to be operational again.

Speaker 1: Mhm.

Speaker 2: That would probably be more interesting to them. Cuz it took like, for I don't know, a few cuz they have to clear the runway of the aircraft, then they have to check for all the small pieces that broke off,

Speaker 1: Yeah.

Speaker 2: and then after that and only then can they patch up the runway with all the divots and potholes or whatever. So, you can mention this as one of the big factors that could save their time in the event of a crash, more quoted F-16 crash.

Speaker 1: Right, got you.

Speaker 2: And one more thing. Tengah Air Base is expanding, so they could

Speaker 1: Yeah.

Speaker 2: Because the whole new runway, the what Lim Chu Kang Road or something. Because it's a whole new runway, since it's a fresh thing, they could look into implementing it from the get-go rather than as an afterthought.

Speaker 1: Right. Do you know they're making a like expanding the air base?

Speaker 2: Yeah, they're even making one at Changi Airport Terminal 5.

Speaker 1: That one's yeah, yeah.

Speaker 2: Yeah, that one's because pilot, but it's shutting down. So, they got to come here.

Speaker 1: Yeah. You got to shift there.

Speaker 2: Yeah. Right.

Speaker 1: All right. If not, that's all the questions I have for today. Thanks for the input. And I really appreciate you taking the time to talk to me today and sharing your experience. Yeah. Appreciate you.

Speaker 2: No worries, man. All the best.

Speaker 1: Yeah, thanks. Thank you, thank you.

Speaker 2: Yeah.

Speaker 1: Bye.

Speaker 2: Bye.

Speaker 1: Thanks, L. Thank you.

Speaker 2: Yep.